

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 – 2024 | Fossil Fuels & Industry

Data: Global Carbon Budget 2025 | NASA / NOAA Mauna Loa Observatory

429 ppm

Atmospheric CO₂ (Feb 2026)

35+ Bt/yr

Global Fossil CO₂ Emissions

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million (Feb 2026)

Pre-industrial baseline: ~280 ppm

+53% increase in ~200 years

Context & Acceleration

- ▶ At least 1.5× greater than the natural increase observed at the end of the last ice age 20,000 years ago
- ▶ That natural change took thousands of years; this one has occurred in roughly two centuries
- ▶ Driven primarily by burning coal, oil, and natural gas
- ▶ Confirmed by both ground-based and satellite measurements

Global Fossil Fuel CO₂ Emissions Grew Six-Fold Since 1950

6 Bt

Annual CO₂ Emissions — 1950



20 Bt

Annual CO₂ Emissions — 1990



35+ Bt

Annual CO₂ Emissions — 2024

Key Trajectory Points

- ▶ Pre-Industrial Revolution: CO₂ emissions were negligible
- ▶ Growth remained relatively slow until mid-20th century
- ▶ Emissions nearly quadrupled from 1950 to 1990
- ▶ **Growth has slowed in recent years, but emissions have not yet peaked**

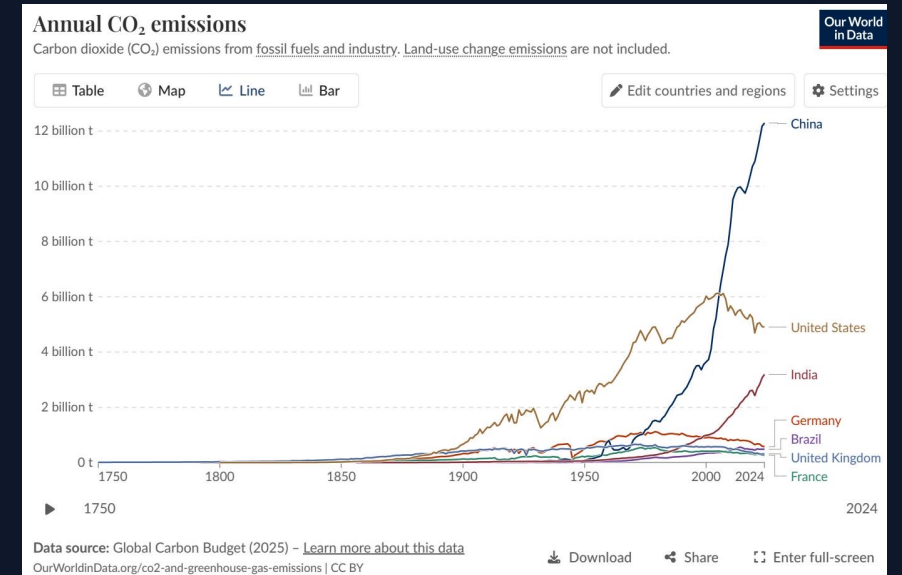
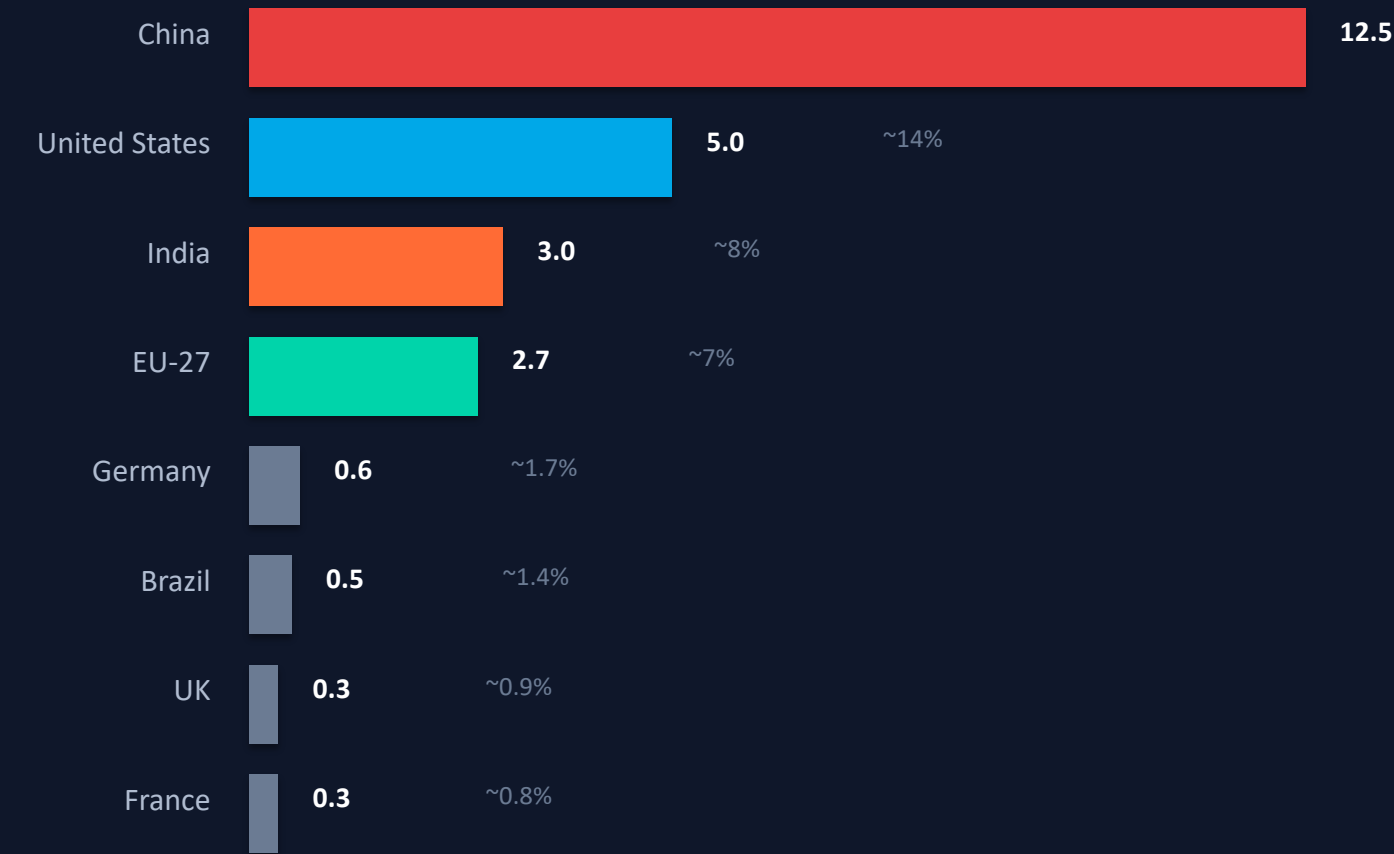


Figure: Annual CO₂ emissions by country

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



Trends (Recent)

- China ~12.5 Bt**
Slight increase, >25% of global total
- United States ~5.0 Bt**
Declining from ~6 Bt peak (c. 2005)
- India ~3.0 Bt**
Rising rapidly, 3rd largest emitter
- EU-27 ~2.7 Bt**
Declining, share fallen substantially

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

Europe + US: >90%

- ▶ Europe and the US dominated global emissions
- ▶ Rest of world: <10%
- ▶ Asia, Africa, South America negligible



2024

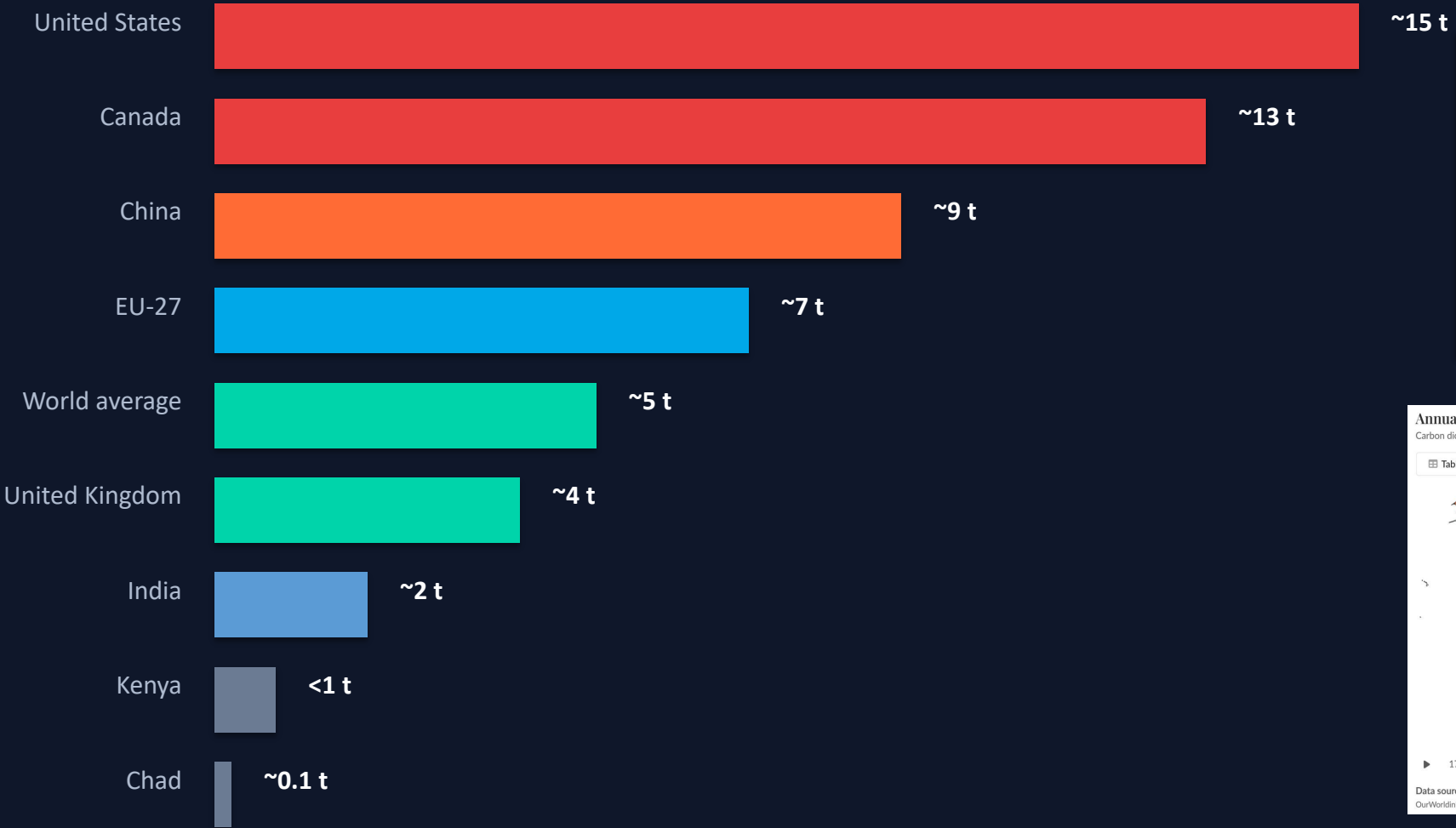
Asia ~50%

US+EU <33%

- ▶ Asia: ~50% (home to ~60% of world population)
- ▶ US + Europe: <33% combined
- ▶ Africa & South America: 3–4% each
- ▶ Similar to international aviation & shipping

By 1950, Europe + US still represented over 85% of global emissions

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



150×
gap between highest and lowest
per capita emitters

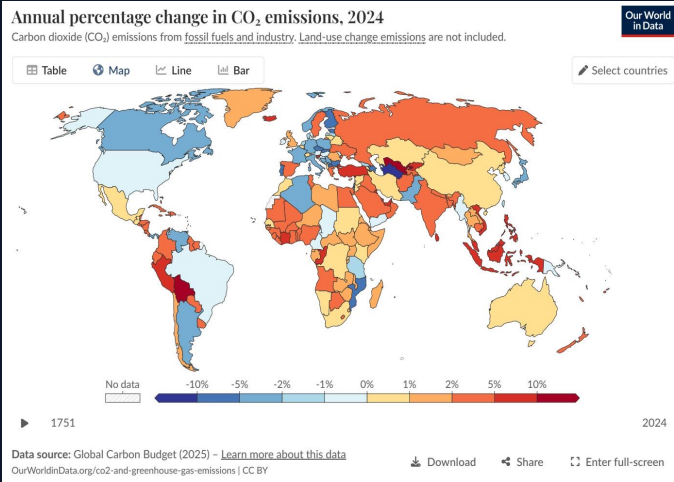


Figure: Per capita CO₂ over time

US and Europe Cut Emissions While Russia and SE Asia Increase

▼ Declining (–1% to –5%)

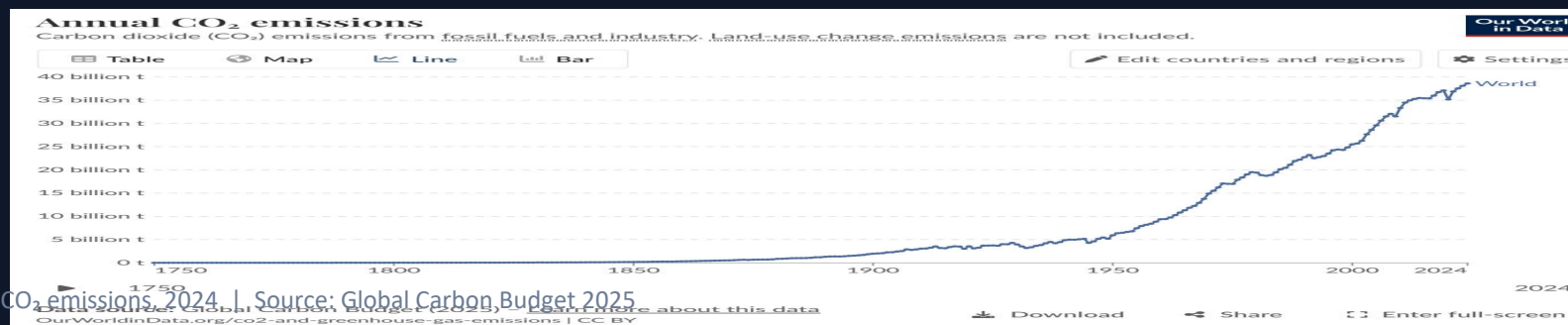
- ▶ United States
- ▶ Much of Europe

Shown in blue on the 2024 annual change map

▲ Increasing (+2% to +10%+)

- ▶ Russia
- ▶ Parts of Southeast Asia
- ▶ Several African nations
- ▶ Some South American countries (>10%)

Shown in orange/red on the map



Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.

	Country	Primary Energy Mix	Per Capita CO ₂
Lower CO ₂	France	High nuclear + renewables	~4 t/person
	United Kingdom	Nuclear + growing renewables	~4 t/person
Higher CO ₂	Germany	Higher fossil fuel share	~8 t/person*
	Netherlands	Higher fossil fuel share	~8 t/person*

Key Insight

Policy and energy mix choices — not just prosperity — determine per capita outcomes. France and the UK achieve similar living standards to Germany and the Netherlands with roughly half the per capita CO₂.

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

01

Global emissions exceed 35 Bt/year and have not yet peaked, despite slowing growth rates.

02

China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the US (~5 Bt); India (~3 Bt) is rising fast.

03

The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today.

04

Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t) from the lowest (Chad at ~0.1 t).

05

Policy and energy mix choices — not just prosperity — determine per capita outcomes (e.g., France vs. Germany).